Below is a detailed Use Case Model for the Tandem t:slim Insulin Pump SImulator. Each use case includes primary actors, Stakeholders,Pre-condition(s),Success Guarantee(s), Main Success Scenario and Extensions:

Use Case Model:
Use Case 1: Power On / Power Off
Primary Actor(s): User
Stakeholders: Tandem t:slim X2 Insulin Pump System
Pre-condition(s): The Pump is charged and connected properly.
Success Guarantee(s): The Pump powers on or off successfully and transitions to the lock screen without any issues.
Main Success Scenario:

1. The user presses and holds the power button.
2. The pump initiates its startup sequence and displays the lock screen.
3. The user enters the correct PIN to unlock the pump (if enabled).
4. The user navigates to the system menu and selects the option to power off.
5. The pump displays a confirmation message and powers down safely.

Extensions:

- E1: The pump does not respond to the power button due to a hardware failure.
- E2: The screen freezes during the boot sequence, requiring a manual reset.
- E3: The lock screen fails to accept the correct PIN due to a software error.
- E4: The shutdown process is interrupted due to a low battery warning.

Use Case 2: Configure Personal Profiles (CRUD)
Primary Actor(s): User
Stakeholders: User
Pre-condition(s):

1. Pump is set up (UC1).
2. User is on the pump's main screen or can navigate to the "Profiles" section.

Success Guarantee(s):

1. New or updated personal profile is stored in the pump.
2. Obsolete profile can be removed.
3. The pump can use the selected profile for insulin delivery.

Main Success Scenario:

1. User navigates to "Personal Profiles."
2. User selects "Create new profile" or "Edit" an existing one.
3. User inputs or changes:
    - Basal rates / time-based segments
    - Carbohydrate ratio
    - Correction factor
    - Target glucose range
4. User names the profile.

5. System saves the profile.
6. (Optional) User can delete an old profile if no longer needed.

Extensions:

- E1: Invalid numeric range (e.g., negative basal rate)
    - System shows an error, user re-enters.
- E2: User cancels changes
    - System discards unsaved updates.

Use Case 3: Deliver Manual Bolus

Primary Actor(s):

- User (the person operating the Tandem t:slim Simulator)

Stakeholders:

- Patient / User: Needs accurate and timely insulin delivery.
- Healthcare Provider: Relies on correct bolus data for evaluating therapy.
- Project Team: Must ensure the simulator accurately processes and logs bolus actions.

Pre-condition(s):

1. At least one personal profile (with carb ratio, correction factor) exists in the simulator.
2. The simulator is powered on and functioning without critical errors.
3. The user can access the “Bolus” menu option from the main screen.

Success Guarantee(s):

- The simulator calculates and delivers the correct bolus amount based on the user’s input and profile settings.
- Bolus delivery is logged in the system history.
- The user receives feedback indicating whether the bolus was successful.

Main success scenario:

1. Navigate to Bolus Screen
    - The user selects “Deliver Bolus” from the home or main menu.
2. Enter Blood Glucose and/or Carb Intake
    - The simulator prompts the user for current BG and carbohydrate amount.
    - The user enters these values (e.g., BG=150 mg/dL, Carbs=50g).
3. System Calculates Suggested Dose
    - Based on the user’s profile (carb ratio, correction factor, insulin on board), the simulator presents a suggested bolus amount (e.g., 4.5 units).
4. Confirm or Adjust Bolus

    - The user reviews the suggestion and can either accept it or manually change it (override).
    - The user confirms the final dose to deliver.
5. Bolus Delivery
    - The simulator processes the bolus delivery, updating the Insulin on Board (IOB) calculation.
    - A notification indicates that the bolus was delivered successfully.
6. Log the Event
    - The system records the date, time, BG/carbs entered, and final bolus amount in the history.
    - The user returns to the main menu.

Extensions:

1. **Insufficient Insulin**
    - If the system detects low reservoir volume before delivery, it warns the user.
    - The user must "refill" (in simulation) or reduce the bolus before proceeding.
2. **User Override**
    - If the user chooses a different dose than suggested, the simulator notes this override in the history.
    - The bolus is still delivered, but the user is reminded that the calculated dose was adjusted.
3. **Error During Delivery**
    - If an occlusion or device error occurs mid-bolus, the simulator suspends delivery and alerts the user.
    - The user addresses the issue (e.g., clear occlusion) before attempting again.
4. **Canceled Bolus**
    - If the user cancels or exits the bolus screen before confirmation, no insulin is delivered.
    - The simulator discards the partial calculation and returns to the main menu.

Use Case 4: Start / Stop / Resume Insulin Delivery
Primary Actor(s):

- User

Stakeholders:
- Patient/User (Needs safe and consistent Insulin delivery)
- Healthcare Provider (May review insulin delivery data for treatment evaluation)
- Project Team (Must ensure accurate simulation of insulin control processes)

Pre-condition(s):
- The simulator is powered on and functioning correctly
- At least one active personal profile with basal rate settings exists

- The system is not currently in an error state (occlusion, critically low battery)

Success Guarantee(s):

- Insulin delivery is started, stopped, or resumed correctly based on the user input or system triggers
- Events (start, stop, resume) are logged appropriately
- The pump reflects the current insulin delivery state accurately on the screen

Main Success Scenario:

1. Start Delivery
    - User navigates to the "Insulin Delivery" or "Basal Settings" menu.
    - User selects "Start Delivery."
    - System checks for an active profile and begins basal insulin delivery at the specified rate.
    - A confirmation message appears indicating delivery has started.
2. Stop Delivery
    - User selects “Stop Delivery” from the same menu.
    - System suspends insulin delivery and logs the stop event.
    - A confirmation message appears showing that delivery has been paused.
3. Resume Delivery
    - User selects “Resume Delivery.”
    - System restores the previously active basal rate (or allows user to select a new one).
    - A confirmation appears, and delivery continues.
    - The system updates status indicators

Extensions:

- E1: No Active Profile
    - If no profile is available, the system prompts the user to create one before starting delivery.
- E2: Low Insulin Warning
    - If insulin volume is low, the system warns the user before starting/resuming delivery.
    - The user may choose to proceed or refill in simulation mode.
- E3: Automatic Stop Triggered by CGM
    - If CGM detects glucose < 3.9 mmol/L, the system automatically suspends delivery.
    - A notification explains the reason and logs the automatic stop.
- E4: Resume Blocked Due to Error
    - If the system detects a device error (occlusion, battery < 5%), it blocks resuming and prompts the user to resolve the issue first.

Use Case 5: View and Interpret Pump Information / History

Primary Actor(s):

- User (the person operating the Tandem t:slim Simulator)

Stakeholders:

- User's Healthcare Provider: May review the history logs for treatment decisions.
- Project/Design Team: Ensures the simulator's history and status information function correctly and are easy to interpret.

Pre-condition(s):

1. The Tandem t:slim Simulator is powered on and functioning normally.
2. The user has already created at least one profile or delivered at least one bolus (so that some history exists).
3. The user is at the main menu or has access to navigate to the history/status section.

Success Guarantee(s):

- The user can view a clear summary of recent pump activities (bolus deliveries, suspensions, error messages, etc.).
- The user can interpret key pump data (battery life, insulin remaining, insulin on board) without confusion.
- The system accurately reflects the latest information and logs all events reliably.

Main Success Scenario:

1. **Navigate to Status/History Section:**
    - The user selects a "Status" or "History" option from the main menu.
2. **Display Pump Status:**
    - The simulator shows current battery level, amount of insulin remaining in the reservoir, and the calculated Insulin on Board (IOB).
    - The user confirms the pump is functioning within normal ranges (e.g., sufficient battery, enough insulin left).
3. **View Event History:**
    - The user sees a chronological list of past actions (e.g., bolus deliveries, suspensions, errors).
    - For each event, the simulator displays date/time, event type (bolus, error, etc.), and brief details (amount delivered, error code, etc.).
4. **Select/Review an Event:**
    - The user can highlight or click on a specific event to view extended details (bolus calculations, overrides, or error messages).
5. **Interpret Findings:**
    - Based on the displayed history, the user may notice patterns (e.g., frequent corrections, battery drain) or confirm correct dosages were delivered.
    - The user can decide whether any adjustments or further actions are needed (refill insulin, charge battery, change profile settings).

6. **Return to Main Menu:**
   - Once finished reviewing, the user exits the status/history screen, returning to the simulator's main interface.

Extensions:

1. **No Historical Data**
   - If there are no prior events, the simulator displays a message such as "No History Found."
   - The user returns to the main menu.
2. **Export/Print Option (If Implemented)**
   - The user selects "Export History" to generate a report file (CSV, PDF, etc.).
   - If the export fails (e.g., invalid file path), the system shows an error message and prompts the user to try again.
3. **Error Alert During Viewing**
   - If, while checking history, the simulator detects a new error (like low battery), an alert pops up.
   - The user must address this alert before returning to review the history.
4. **Partial or Corrupted Log**
   - If a previous crash or power loss occurred, the simulator may show a partial event log or a warning (e.g., "Some records may be missing due to an unexpected shutdown").
   - The user can still view whatever data remains and proceed normally.

Use Case 6: Respond to Errors and Malfunctions
Primary Actor(s): User
Stakeholders:
- Tandem System - Maintain integrity and user safety
- Support team - May assist user in resolving issues
- User - Must be able to respond safely to alerts

Pre-condition(s):
- Pump is in an error or alert state (e.g., occlusion, empty cartridge, low battery)
- User is notified via visual, tactile, or audible alarm

Success Guarantee(s):
- The user is alerted about the problem
- The issue is clearly described and resolvable by user or support
- Logging occurs automatically

Main Success Scenario:
1. Error or malfunction occurs
2. Pump displays the alert message with resolution instructions
3. User acknowledges the alert
4. If it is a fixable error, user performs required action

5. System checks and confirms the issue is resolved
6. Pump resumes normal operation

Extensions:

- E1: Error persists beyond fix, user is prompted to contact Tandem support
- E2: Battery dies during error - Pump must be charged and restarted; log may be recovered
- E3: Multiple errors - System queues alerts based on severity

Use Case 7: Simulate CGM Feedback
Primary Actor(s):
Stakeholders:
Pre-condition(s):
Success Guarantee(s):
Main Success Scenario:
Extensions:

Use Case 8: View Guide / Information
Primary Actor(s): User
Stakeholders: User
Pre-condition(s):

1. Pump is powered on and at the home screen (or user can navigate there).
2. A “Guide” or “Help” button or menu option is available.

Success Guarantee(s):

1. User sees textual or graphical instructions for core pump features (profiles, manual bolus, error handling, etc.).
2. System logs that the user accessed the help section.
3. User can close the guide and return to normal operation at any point.

Main Success Scenario:

1. User taps “Guide” / “Help” on the home screen or options menu.
2. System displays an indexed or scrollable help section covering the main features (e.g., “Setting Up Personal Profiles,” “Manual Bolus,” “Error Alerts”).
3. User navigates within the guide to read specific topics.
4. User exits the guide and returns to the pump’s previous screen.

Extensions:

Traceability matrix

Req ID	Requirement (from Feature List)	Related Scenario / Use-Case	Key Design Element(s) (class / UI / method)
R1	Simulated clock shall advance 1 real s = 5 sim-min and drive all time-based components.	Time/CGM simulation sequence diagram	`SimulationClock` class (tick()), connections in `mainWindow` constructor, every page that calls setSimulationClock()
R2	CGM readings shall be generated every 5 sim-min and plotted for 1 h / 3 h / 6 h windows.	«CGM sampling» sequence	`BGSimulator::onTick()`; `GraphPage` live plot; `GraphPage` filter buttons
R3	Manual bolus delivery with user-entered BG + carbs shall compute dose using ICR & CF and subtract IOB.	"Manual bolus" sequence diagram	`BolusPage` UI → `Pump::deliverBolus()` → `BolusCalculator`
R4	Profile management: create, edit, delete, activate ≥ 3 personal profiles.	"Manage personal profiles" sequence	`ProfileManager` class; `ProfileListPage`, `ProfileEditorPage`, signals in `mainWindow`
R5	Selecting a profile shall update active ICR & CF for all future bolus and BG calculations.	«Select profile»	`Pump::selectActiveProfile();` `BGSimulator::setProfile()` connection
R6	Pump shall log all events (CGM, insulin, warnings) with simulated timestamp to a history page.	«View history log»	`Pump::logEvent()`; `HistoryLog` struct; `HistoryLogPage` table + `addEntry()`
R7	When insulin cartridge < 30 u the system shall raise *Low Insulin* warning and show	«Low-insulin path»	`Pump::checkLevels()` ➔ `ErrorHandler::raise(LowInsulin)` ➔ `WarningDialog` ('Fill insulin' button)

	modal dialog with “Fill insulin”.		
R8	When battery < 20 % the system shall raise *Low Battery* warning and allow “Charge battery”.	«Low‑battery path»	`Pump::checkLevels()` ➔ `ErrorHandler::raise(LowBattery)` ➔ `WarningDialog` (‘Charge battery’)
R9	If BG predicted > maximum (13 mmol/L) the pump shall deliver automatic correction bolus (Control‑IQ).	“Automatic correction” scenario	`BGSimulator` → signal to `Pump::deliverBolus()` with auto flag
R10	If BG predicted < minimum (4 mmol/L) the pump shall stop insulin (*Emergency Stop*).	«Hypoglycemia safety»	`BGSimulator` → `Pump::emergencyStop()` & `ErrorHandler::raise(BGLow)`
R11	The system shall allow refilling insulin and charging battery from Home screen.	Basic maintenance scenario	Buttons in `HomePage` & `SettingsPage`; `Pump::fillInsulin()`, `Pump::chargeBattery()`
R12	The application shall start locked and require 4‑digit PIN to unlock Home page.	«Unlock device»	`LockPage` UI; signal `authenticated` → `mainWindow::onActionHome()`